

PRASHASTI SATPATHI

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Technology(Health Informatics)

CGPA: 8.03

Bhopal, Madhya Pradesh

2023-2027

Shiv Jyoti Senior Secondary School

HSC

75.4%

Kota, Rajasthan

2021-2022

SKILLS

Languages:	C/C++, Python, Java, SQL, JavaScript, TypeScript, Bash/Shell Scripting
Frameworks:	ReactJS, ExpressJS, NodeJS, BunJS, QtC++, Kafka
Technologies:	Front-End & Back-End Development, UI/UX Design, Cloud Computing, Cybersecurity, DevOps, CI/CD, Git, Docker, Kubernetes
Technical Subjects:	System Design, OOPs, Data Structures & Algorithms, DBMS, Operating Systems, Networking, Computer Architecture, Software Engineering Principles
Soft Skills:	Problem Solving, Analytical Thinking, Decision Making, Communication, Leadership, Time Management, Mentorship, Public Speaking, Collaboration

PROJECTS

Secure Hybrid RAG Engine

- Engineered a **secure Hybrid RAG system** using FastAPI and LangChain to automate knowledge retrieval across **1,000+ pages of complex plant engineering layouts and maintenance schemas**.
- Maintained **100% data confidentiality** within the enterprise network **by architecting a dual-pipeline fallback utilizing Pinecone vector spaces and a localized PyMuPDF disk execution perimeter**.
- Accelerated API response times to sub-second levels by migrating sequential, blocking network requests into non-blocking parallel workflows using AsyncGroq and asyncio.gather.
- Replaced manual document search times with **instant query resolution, utilizing LlamaParse layout-aware structural extraction** to deliver precise context blocks to internal engineering teams.

Codebasic: A programming language for beginners | GitHub

- Developed a simple programming language **interpreter/compiler from scratch using TypeScript/JavaScript**, implementing lexical analysis, syntax parsing, and semantic analysis.
- Created a beginner-friendly language with support for variable declaration, arithmetic operations, conditional statements (if/else), and a show/print statement for output — helping novices understand programming concepts.
- Gained **hands-on experience with AST (abstract syntax tree)** design, traversal, and interpreter semantics — improving understanding of compiler/translator architecture and language design.

Real-Time Healthcare Event Processing Platform | GitHub

- Designed and developed a distributed platform for **real-time processing of chronic disease health data** using **Apache Kafka** for ingestion and **Siddhi CEP** for complex event processing.
- Derived clinical decision rules from datasets using **Random Forest algorithm(84% accuracy)** and applied them via Siddhi CEP to detect events such as **stroke risk, cardiac alerts, and diabetes crises**.
- Processed **1500+ complex events per second** with low-latency alert generation and evaluated the system for scalability, performance, and accuracy.

EXPERIENCE

- Gen AI Intern at Reliance Power Limited:** Built a secure hybrid RAG system with FastAPI/LangChain for 1000+ pages of plant engineering layouts, using Pinecone-PyMuPDF fallback pipelines and AsyncGroq to improve API throughput by 40% with sub-second responses.
- Research Intern at IISD:** Automated SDLC data collection and collaborated with remote teams to streamline research delivery.

CERTIFICATIONS

Applied Machine Learning in Python	University of Michigan	Mar 2025
Regulatory Requirements for Clinical Trials	NPTEL	Sep 2025
Complete Data Science, ML, DL, NLP Bootcamp	Udemy	Dec 2024

CO-CURRICULAR ACTIVITIES

Operations Team Co-Lead at Dishari Club, VIT Bhopal: Managed operational logistics for events with 550+ attendees, requiring high-level verbal coordination.